

FRUIT AND VEG CLASSIFICATION

CNN PROJECT

PURPOSE OF THE PROJECT

This project purposes to use a Convolutional Neural Network (CNN) to classify fruits and vegetables. The project is a great opportunity to improve the ability to extract information from visual data using image processing and deep learning techniques.

Fruit and vegetable classification has an important place in the food industry, agriculture and retail. This project aims to provide automatic identification of fruits and vegetables by training and testing a CNN model.



FRUIT AND VEG SPECIES

"Corn", "Cabbage", "Cauliflower", "Capsicum",
"Cucumber", "Carrot", "Banana", "Turnip",
"Kiwi", "Beetroot", "Raddish", "Grapes",
"Lemon", "Potato", "Spinach", "Pear",
"Sweetpotato", "Orange", "Eggplant",
"Jalepeno", "Paprika", "Watermelon", "Chilli
pepper", "Onion", "Tomato", "Peas", "Bell
pepper", "Soy beans", "Garlic", "Lettuce",
"Apple", "Mango", "Pomegranate",
"Sweetcorn", "Ginger", "Pineapple"



WHAT WE DID IN THE PROJECT?



We classified the fruit and veg species according to their classes. There were 36 different species. There were high quality photos for each fruit and veg species. Each image was labeled with the relevant species.



We applied various pre-processing steps to make the dataset suitable for training. We reduced all the images to a certain size (32x32 pixels). We scaled the pixel values to a certain range (255.0)



MODELING



We took normalized images of a certain size (32x32 pixels) as input. We used a series of convolution layers. We used activation functions like ReLU. We added max pooling layers. We gave 36 neurons to the last layer since we have 36 classes. We used the softmax function in the last layer.



We divided the dataset into 80% training and 20% testing. The training set is used to learn the model, while the testing set is used to evaluate the overall performance of the model. During the training process, the loss function is monitored. Here we used sparse categorical crossentropy.

TRANSFER LEARNING

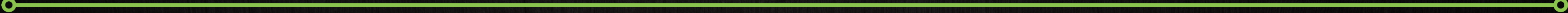
VGG16 is a popular deep learning architecture, often used in image classification tasks. We scaled the data in the Indian bird data and divided it into training and test data. We used the layers of VGG16. We determined our epoch number. We trained our model.



CREATING A WEBSITE

We developed a website using Streamlit with the model we saved. We uploaded it to Hugging Face. So other people can access our site and we finished our project...





THANKS

